

Miranda Caputo

✉ miranda.caputo@rockets.utoledo.edu

🆔 0000-0002-2957-3924

🔗 m-caputo.github.io/mcaputo/

EDUCATION

University of Toledo

2020 - Summer 2026 (expected)

Ph.D. Candidate, Astrophysics

Toledo, OH

Untangling the Formation and Dissolution of Optically Bright and Obscured Stellar Clusters in Nearby Extreme Environments

Thesis Advisor: *Prof. Rupali Chandar*

Santa Clara University

2014 - 2018

Bachelor of Science, Physics

Santa Clara, CA

EXPERIENCE

Graduate Research Assistant Advisor: Prof. Rupali Chandar

2020 - present

University of Toledo

Toledo, OH

Research Assistant Advisor: Dr. B-G Andersson

2019 - 2020

SOFIA Science Center/USRA

Mountain View, CA

Summer Student Advisors: Dr. B-G Andersson & Dr. Kristin Kulas

Summer 2018

SOFIA Science Center/Santa Clara University

Mountain View, CA

Undergraduate Teaching Assistant

2016 - 2018

Santa Clara University

Santa Clara, CA

HONORS AND AWARDS

Physics & Astronomy Outstanding Department Service Award

2023 - 2024

University of Toledo

Astronomy Seminar Best Graduate Student Presentation

Spring 2022

University of Toledo

Clare Boothe Luce Scholar

Summer 2018

Santa Clara University

SUCCESSFUL OBSERVING PROPOSALS

JWST Cycle 3: (P.I.) [GO-06035](#): Hidden Clusters No More! Completing a Full Cluster Catalog from NUV to MIR in *The Extreme* Galaxy NGC 1614 - \$67,350

JWST Cycle 4: (Co.I.) [GO-08564](#): Staring at the Void: Supernova-Driven Star Formation and Suppression in the Largest Local Superbubble

JWST Cycle 2: (Co.I.) [GO-03348](#): Crouching Galaxy, Hidden Stars: Observational Tests of Cluster Formation Theories

ALMA Cycle 11: (Co.I.) 2024.1.01031.S: Resolving the ionized gas from nascent massive star clusters in the Antennae

MENTORING EXPERIENCE

REU Students:

Summers 2022 - 2025

Camille Annan (Indiana University - Bloomington), Jacob Eiermann* (University of Georgia), Aimee Toscano* (Washington & Jefferson College) & Jaidyn Conyer* (University of Akron)
Toledo, OH
Resulted in 2 publications: Eiermann et al. (2025) and Chandar et al. (2023)

Post-Bachelors Student:

2021 - 2023

Yifan Zhang - Current graduate student at Florida State University
Toledo, OH

Undergraduate Chandar Cluster Research Pod:

Fall 2023 - present

10 current and former undergraduate students
Toledo, OH
3 now in graduate programs: Thomas Behling, Tyler Robbins, and Tessa Frinche

High School Students:

Summers 2022 & 2024

RISE program: Kate Wu, Nithya Rajan, Anthony Clark, & Amogh Sharma (24),
Toledo, OH
Max Chandar-Kouba, Jasjeev Singh, & Andrew Zheng (22)

SOFIA Science Center Summer Students:

Summers 2019 & 2020

Sydney Fox-Middleton, Remy Gurarie (Dennis)*, & Joshua Yaldae
Mountain View, CA
Resulted in publication: Caputo et al. (2023)

* denotes students with published work from our collaboration

TEACHING EXPERIENCE

Head Lab T.A.: University of Toledo

2022 - 2023

Introduction to Astronomy Lab
Toledo, OH

Lab T.A.: University of Toledo

2020 - 2021

Introduction to Astronomy Lab
Toledo, OH

Undergraduate Lab T.A.: Santa Clara University

2016 - 2018

Calculus & non-calc. based Mechanics, Optics, & E&M Labs
Santa Clara, CA

SERVICE

Astronomy Journal Club Co-Host

Fall 2023 - present

University of Toledo
Toledo, OH

Department of Physics & Astronomy Graduate Mentor

2023 - present

University of Toledo
Toledo, OH

Physics & Astronomy Python Workshop Presenter

Summers 2022 - 2024

University of Toledo
Toledo, OH

SKILLS

Coding: Proficient with Python. Intermediate Matlab, IRAF, and perl/sh scripts. Introductory level C++, CAD, and LabView. Work with SAO Image DS9 and CARTA.

Instrument Experience: JWST: NIRCам, MIRI, & NIRSpec, HST: WFC3, ACS, & WFP2, SOFIA: upGREAT & HAWC+, Lick Observatory Kast Spectrograph

Specialized Lab Equipment: liquid nitrogen, and lasers

PEER-REVIEWED PUBLICATIONS

1. *NUV-NIR Complete Cluster Demographics and Star Formation Histories in the Antennae*
Caputo, M. (in preparation)
2. *Uncovering Optically Obscured Clusters With JWST NIRCам in Extreme Galaxy NGC 1614*
Caputo, M. (in preparation)
3. *Nowhere Left to Hide: Uncovering All of the Massive Young Embedded Star Clusters in the Antennae with JWST*
Chandar, R., **Caputo, M.**, Goudfrooij, P., et al. (accepted)
4. *The PHANGS-HST-H α Survey: Warm Ionized Gas Physics at High Angular Resolution in Nearby Galaxies with the Hubble Space Telescope*
Chandar, R., Barnes, A. T., Thilker, D. A., **Caputo M.**, Floyd, M. R., Leroy, A. K., Úbeda, L., et al. (2025) [AJ 169, 150](#).
5. *Clusters, Clumps, Dust, and Gas (CCDG) in NGC 1614: Benchmarking Cluster Demographics in Extreme Systems*
Caputo M., Chandar, R., Mok, A., Linden, S., Goudfrooij, P., & Whitmore, B. C. (2024) [AJ 168, 259](#).
6. *A Tale of Three Dwarfs: Cluster-based Star Formation Histories of Blue Compact Dwarf Galaxies*
Chandar, R., **Caputo M.**, Mok, A., Linden, S., Whitmore, B. C., Goudfrooij, P., Cook, D. O., et al. (2024) [ApJ 965, 95](#)
7. *The 3D geometry of reflection nebulae IC 59 and IC 63 with their illuminating star gamma Cas*
Eiermann, J. M.*, **Caputo M.**, T. S.-Y.Lai, & Witt, A. N. (2024) [MNRAS 529, 1680–1687](#).
8. *Physics and Chemistry of Radiation Driven Cloud Evolution. [C II] Kinematics of IC 59, and IC 63*
Caputo M., Soam, A., Andersson, B-G., Dennis, R.*, Chambers, E., Güsten, R., Knee, L. B. G., et al. (2023) [AJ 50, 140](#).
9. *A Tale of Three Dwarfs: No Extreme Cluster Formation in Extreme Star-forming Galaxies*
Chandar, R., **Caputo M.**, Mok, A., Linden, S., Whitmore, B. C., Toscano, A.*, Conyer, J.*, et al. (2023) [ApJ 949, 116](#).
10. *Interstellar Extinction, Polarization, and Grain Alignment in the Sh 2-185 (IC 59 and IC 63) Region*
Soam, A., Andersson, B-G., Straizys, V., **Caputo M.**, Kazlauskas, A., Boyle, R. P., Janusz, R., et al. (2021) [AJ 161, 149](#).
11. *Probing Interstellar Grain Growth Through Polarimetry in the Taurus Cloud Complex*
Vaillancourt, J. E., Andersson, B-G., Clemens, D. P., Piirola, V., Hoang, T., Becklin, E. E., and **Caputo, M.** (2020) [AJ 905, 157](#).

TALKS AND POSTERS

1. **Dissertation Talk: 247th AAS** **2026**
The Formation and Dissolution of Star Clusters in Extreme Star-Forming Galaxies Phoenix, AZ
2. **Poster & Flash Talk: Bridging Scales: star clusters and their host galaxies from the local to high-z Universe** **2025**
The New Frontier of Cluster Detection: Analyzing Cluster Mass Functions of 5 Extreme Galaxies with JWST and HST Matera, Italy
3. **Public Talk: PlaneWave's Annual Open House** **2024**
A Deeper Understanding of Galaxies with JWST Adrian, MI
4. **Public Talk: University of Toledo Physics and Astronomy Colloquium** **2024**
UToledo Research with NASA's Most Powerful Space Telescope Toledo, OH
5. **Talk: Ritter Research Symposium** **2022, 2023, & 2024**
An Introduction to CCDG and the Formation and Destruction of Clusters in Extreme Galaxies Toledo, OH
6. **iPoster: 241st AAS Meeting** **2023**
Cluster Populations in Extreme Galaxy NGC 1614 Seattle, WA
7. **Talk: UToledo Galaxy Group** **2022**
Formation and Destruction of Clusters in the Extreme Galaxy NGC 1614 and Star Formation Histories of Arp 220 and 3 Blue Compact Dwarfs Toledo, OH

8. **Talk: SOFIA Staff Science Symposium** **2019**
Grain Alignment Around the Carbon-Rich AGB Star IRC+10216 Gilroy, CA
9. **Talk: Midwest Magnetic Fields Meeting** **2019**
Optical Polarimetry of AGB star Circumstellar Envelopes Madison, WI
10. **Poster: 233rd AAS Meeting** **2019**
Optical Polarimetry of AGB star Circumstellar Envelopes Seattle, WA
11. **Talk: SOFIA Science Center Seminar** **2018**
Polarization of Optical Light As Seen Through Starless Cloud L182 Mountain View, CA
12. **iPoster: 233rd AAS Meeting** **2018**
Understating Polarization in the Interstellar Medium Through the Theory of Radiative Torque Alignment Denver, CO